Finance GraphRAG - Research Project Plan

Duration: about 26 project weeks at 15-20 hours per week. The schedule assumes the existing Finance-GraphRAG repository and current Stage 1 setup work continue rather than restarting from zero.

At the time of this revision: Stage 1 is the active implementation target: construct an XBRL-derived graph on the same company/filing scope as FinReflectKG and establish a fair graph-level and QA-level comparison.

The scope rule

The core paper is Stages 1-3. Stages 4-5 are extensions, not prerequisites. Do not delay the core evaluation in order to build a larger agent system. Every added component must answer a distinct research question and must be ablated.

The thesis you are aiming at

Write this at the top of the repository README. Every experiment should either support, refine, or falsify it.

> **Authoritative financial structure can improve both retrieval and verification. Instead of asking an LLM to reconstruct financial relationships that already exist in SEC XBRL, we use XBRL as an authoritative graph index for structured financial facts and as a deterministic verification layer for numerical reasoning.**

This is deliberately broader than a simple "XBRL beats FinReflectKG" claim. The project first tests whether XBRL can replace LLM-built graph construction for the structured, numeric portion of financial QA. It then tests whether XBRL remains useful even when an LLM-built graph or text retriever is still necessary.

The central distinction

Layer	What the LLM may do	What XBRL can contribute
Retrieval structure	Construct entities/relations from filing text	Provide concepts, facts, periods, dimensions, presentation/calculation relationships
Multi-hop evidence	Traverse extracted graph relations	Traverse company-concept-period structure and compare across years/companies
Reasoning	Synthesize and calculate from retrieved evidence	Supply exact operands, units, periods, and structural constraints

Layer	What the LLM may do	What XBRL can contribute
Verification	Judge whether an answer is plausible	Deterministically recompute arithmetic and check accounting/period/unit consistency
Narrative coverage	Represent risks, strategy, management discussion	Limited: XBRL does not replace narrative text retrieval

What the paper should not claim

- **Not "XBRL is free."** Parsing, storage, indexing, and engineering still cost compute and time. The defensible claim is zero LLM graph-construction calls for the XBRL portion.
- **Not "XBRL replaces financial RAG."** It does not cover most narrative prose, qualitative reasoning, or every non-GAAP metric consistently.
- **Not "graphs always beat flat retrieval."** Single-fact lookup may remain a flat-RAG strength. The interesting claim is conditional: structure should matter more as relational complexity increases.
- **Not "a correct answer proves retrieval worked."** A model can memorize public-company facts. Traceable correctness and evidence recall must be measured separately.

Research questions and contribution hierarchy

Treat these as an ordered hierarchy. RQ1-RQ3 define the minimum viable paper. RQ4-RQ5 are contingent extensions.

RQ	Question	Status in paper	Decision criterion
RQ1	Can an XBRL-derived graph compete with an LLM-built financial KG for XBRL-answerable financial questions?	Core	Comparable or better evidence retrieval / QA on aligned coverage, with lower LLM construction cost
RQ2	Does the value of authoritative structure increase as questions become more multi-hop or structurally compositional?	Core	Performance advantage grows or remains concentrated in inter-year, cross-company, aggregation, or higher-hop classes
RQ3	Does deterministic XBRL-backed auditing improve reliability after retrieval?	Core	Auditor catches real inconsistencies with low false-positive rate and improves traceable correctness after recovery/re-retrieval
RQ4	Can XBRL verify or repair the structured portions of an LLM-built financial graph?	Extension	Audited FinReflectKG reduces edge/fact errors or downstream QA failures without destroying useful narrative coverage
RQ5	What end-to-end hybrid architecture best combines XBRL, text retrieval, and LLM-created structure?	Extension	Router/hybrid system improves mixed benchmark coverage after the core mechanism is already established

Contribution priority

1. **Authoritative graph indexing.** For structured reported facts, financial graph construction can reuse regulator-maintained XBRL structure rather than re-extracting the same relationships with an LLM.
2. **Structural multi-hop retrieval.** The project tests where graph structure changes retrieval outcomes, not merely whether one system has a higher overall benchmark score.
3. **Deterministic verification.** A subset of financial errors can be checked exactly using arithmetic, period/unit constraints, and accounting relationships.
4. **LLM-graph auditing.** If the extension succeeds, XBRL becomes a constraint layer rather than a wholesale replacement for LLM-created graphs.
5. **Hybrid coverage.** Narrative text remains necessary; the final architecture should route questions according to what structure can actually answer.

Primary hypotheses

- H1: On questions whose gold evidence is representable in XBRL, an XBRL-derived graph will retrieve at least as reliably as FinReflectKG on aligned facts/relations while requiring no LLM calls to construct that structured graph.
- H2: The relative value of the XBRL graph will be largest on inter-year, cross-company, aggregation, and other multi-hop structural questions rather than on single-fact lookups.
- H3: Deterministic verification will convert some retrieval/generation failures into recoverable failures by detecting inconsistent values, periods, units, or arithmetic.
- H4 (extension): XBRL constraints will improve the structured factual reliability of an LLM-built KG while leaving narrative relations available for non-XBRL questions.

The five-stage experimental map

The stages are cumulative, but only Stages 1-3 are required to tell a complete research story.

Stage	Main comparison	What changes	What it isolates
1. XBRL graph vs FinReflectKG	Same companies/filings, same question slice	Graph source	Whether authoritative XBRL can substitute for LLM-built structure
2. Multi-hop stress test	Flat RAG vs FinReflectKG vs XBRL graph	Hop count / structural relation type	Where graph structure actually matters
3. Deterministic audit	XBRL graph vs XBRL graph + auditor	Verification layer	Whether exact checking improves reliability independently of retrieval
4. Audit LLM-built graph	FinReflectKG vs FinReflectKG + XBRL constraints	Verification/repair of LLM graph	Whether XBRL is useful even when LLM-created structure remains
5. Hybrid end-to-end system	Best baselines vs router + XBRL + text (+ optional KG)	System composition	Practical coverage across numeric, structural, narrative, and mixed questions

Why multi-hop is not a separate thesis

Multi-hop is the stress test applied across the systems. It tells you where structural retrieval should help. FinReflectKG-MultiHop therefore belongs mainly in Stage 2 as an evaluation resource, rather than becoming a separate project branch.

Critical path versus stretch path

Path	Required components	Paper-level result
Minimum viable paper	Baselines + Stage 1 + Stage 2 + graph/evidence metrics	Authoritative XBRL graph as a retrieval structure; multi-hop structural analysis
Preferred core paper	Minimum + Stage 3 deterministic auditor	Retrieval plus deterministic verification, with cost/traceability analysis
Stretch paper	Preferred core + Stage 4 and/or Stage 5	XBRL as a constraint layer for LLM-built graphs and/or a full hybrid financial RAG architecture

Phase 0 - Shared setup before the stages (Weeks 1-2)

Do this once. Every later stage should reuse the same normalized data layer, evaluation harness, caching, and question taxonomy.

0.1 Lock the aligned experimental slice

- **Company/filing overlap first.** Identify the exact companies and filing years covered by the FinReflectKG data you downloaded. Build the first XBRL graph only on that overlapping slice so Stage 1 is a fair comparison.
- **Freeze a development slice.** Use a small fixed set for iteration. Do not repeatedly tune on the final test questions.
- **Keep a held-out test slice.** The final comparison should be run after graph construction and traversal parameters are frozen.
- **Store provenance.** Every XBRL fact should keep concept, entity/CIK, filing/accession, period, unit, dimensions, value, and source filing.

0.2 Normalize a common evidence schema

The systems do not need identical internal graphs, but the evaluator needs a common representation of retrieved evidence.

Field	Purpose
entity/company	Allows company-level alignment and cross-company evaluation
concept/relation	Allows concept-level evidence matching between XBRL and KG nodes/edges
period	Prevents silent fiscal/calendar-year mistakes and supports inter-year hops

Field	Purpose
value / text span	Supports exact numerical evidence checks and narrative evidence
unit / dimensions	Supports unit and segment/dimensional auditing
source filing	Enables traceable correctness and citation checks
system provenance	Records whether the evidence came from flat RAG, FinReflectKG, XBRL traversal, or a hybrid path

0.3 Shared baselines

1. **No retrieval.** Measure backbone-model memorization, especially for large public companies.
2. **Flat RAG.** Dense or hybrid chunk retrieval over filing text, with the same generator used for the graph systems.
3. **FinReflectKG.** Use the downloaded LLM-built financial KG as the independent graph baseline rather than building a deliberately weaker one yourself.
4. **XBRL graph.** The main system under study, added only after the baseline harness is stable.

0.4 Shared metrics

Metric family	Metrics to report
Answer quality	Exact match / normalized numeric accuracy; tolerance-aware numeric correctness where appropriate
Evidence quality	Evidence Recall@k, evidence precision when gold evidence exists, source/filing correctness
Traceability	Traceable Correct, Untraceable Correct, Traceable Incorrect
Graph coverage	Question answerability by XBRL; concept mapping success; aligned node/edge coverage
Multi-hop	Accuracy by hop count and structural type; complete-path evidence recall
Audit	Detection precision/recall, false-positive rate, recovery success after re-retrieval
Efficiency	Index build wall time, LLM construction tokens/cost, online latency, online tokens

Checkpoint 0

You pass when one command can evaluate no-retrieval and flat RAG on a fixed dev slice; the FinReflectKG overlap set is known; XBRL facts can be represented in the common evidence schema; and every retrieval/LLM call is cached by input hash.

Stage 1 - XBRL graph versus FinReflectKG (Weeks 3-7)

Research question: can regulator-maintained XBRL structure compete with an LLM-built financial knowledge graph on the portion of financial QA that XBRL can represent?

1.1 Build the XBRL fact graph

Start with the smallest graph that can support a fair comparison. Do not build the final hybrid system yet.

- **Fact nodes or records.** (company, concept, period, dimensions, unit, value, filing).
- **Concept structure.** Presentation parent/child relationships and, where reliably obtainable, calculation relationships with weights.
- **Temporal edges.** Same company + same normalized concept across periods.
- **Cross-company alignment.** Same normalized concept + comparable period across companies.
- **Filing provenance.** Every retrieved fact must be traceable to its filing/accession.

1.2 Alignment to FinReflectKG

The hardest part of Stage 1 is not graph traversal; it is defining what counts as a fair overlap.

Alignment question	Rule
Same companies?	Restrict the primary comparison to company overlap first.
Same years/filings?	Restrict to filing-year overlap or report coverage separately.
Same concepts?	Build a concept/alias mapping only for concepts needed by the evaluation questions; log unmapped cases.
Narrative-only relations?	Mark as outside XBRL coverage rather than scoring them as XBRL retrieval failures.
Company-specific extension tags?	Normalize cautiously; preserve raw tag and mapping confidence.
Restated values?	Treat filing provenance and period as part of fact identity; do not collapse conflicting-but-valid restatements silently.

1.3 Two evaluations inside Stage 1

1. **Graph/evidence quality.** Before final QA, compare whether each graph contains and retrieves the evidence needed for the aligned questions.
2. **Downstream QA.** Feed retrieved evidence to the same generator and measure answer correctness plus traceability.

1.4 Optional negative-result analysis

Use REFinD or another suitable relation ground truth only where its schema can be aligned fairly. The goal is not to attack FinReflectKG; the goal is to quantify which classes of LLM-extracted relation error propagate into retrieval or QA.

- Missing relation or fact
- Wrong entity/value attachment
- Direction error
- Temporal mismatch
- Duplicate/conflicting relation
- Correct relation but not retrievable from the query seed

Stage 1 primary table

System	XBRL-answerable coverage	Evidence Recall@k	QA accuracy	Traceable Correct	Build LLM tokens
FinReflectKG					
XBRL graph					0 for XBRL graph construction
Flat RAG	n/a				0 graph-construction tokens

Stage 1 go/no-go rule

Proceed to make XBRL the retrieval centerpiece if it is competitive on aligned XBRL-answerable evidence and shows a clear construction-cost / fidelity advantage. If it underperforms, diagnose whether the cause is concept mapping, traversal, coverage, or a genuinely weaker retrieval representation before moving on.

Stage 2 - Multi-hop structural stress test (Weeks 8-11)

Research question: does the usefulness of authoritative structure increase as the number or type of required relations becomes more complex?

2.1 The evaluation taxonomy

Class	Example requirement	Expected retrieval challenge
Single-fact	One company, one concept, one period	Flat retrieval may be sufficient
Intra-document composition	Two or more values from one filing	Evidence completeness matters more than similarity alone
Inter-year	Same company/concept across periods	Requires temporal linking and period normalization
Cross-company	Same concept/period across companies	Requires entity alignment and comparable periods
Aggregation / set	Filter or aggregate many companies/facts	No single chunk contains the answer
Mixed structural + narrative	Numeric comparison plus management explanation	Requires hybrid evidence; reserve mainly for Stage 5

2.2 Benchmark roles

- **FinReflectKG-MultiHop.** Primary structural benchmark if the available question/evidence set is usable. Preserve its intra-document, inter-year, cross-company, and hop-count labels where available.
- **FinanceBench open subset.** Use for natural expert-written questions and for comparability with prior financial retrieval work, but do not rely on 150 questions alone.
- **XBRL-generated multi-hop supplement.** Generate large exact-answer question sets from authoritative facts for statistical power. Hand-validate a sample and clearly label them templated.

- **CRAG finance or other larger sets.** Use only where the retrieval setting and evidence can be aligned cleanly; do not expand benchmark count just for appearance.

2.3 What "multi-hop success" means

A correct final number is not enough. For a k-hop question, record whether the complete evidence path was retrieved.

Outcome	Interpretation
Correct answer + complete path	True structural retrieval success
Correct answer + incomplete path	Potential memorization, guessing, or unsupported reasoning
Wrong answer + complete path	Generator/reasoning failure
Wrong answer + incomplete path	Retrieval/path failure
Correct operands + wrong period/unit	Evidence-selection failure that may be catchable by Stage 3 auditing

Stage 2 headline figure

System	1-hop	2-hop	3-hop	Inter-year	Cross-company	Aggregation
Flat RAG						
FinReflectKG						
XBRL graph						

Stage 2 success condition

The strongest result is not necessarily a large Overall gain. A publishable structural result can be: flat retrieval is competitive on single-fact questions, while the XBRL graph wins sharply on specific multi-hop classes. The paper should explain the mechanism behind the column-level differences.

Stage 3 - Deterministic XBRL-backed auditing (Weeks 12-15)

Research question: after retrieval, can authoritative structure detect and help recover from numerical or structural mistakes before an answer is finalized?

3.1 Separate retrieval from verification

The auditor is deliberately downstream of retrieval. This allows a clean ablation: XBRL graph alone versus the same graph plus exact checks.

3.2 Auditor checks, in implementation order

Check	Inputs	Failure it can detect	Priority
Endpoint arithmetic	Retrieved operands + claimed derived answer	Wrong growth rate, margin, ratio, difference, sum	Highest

Check	Inputs	Failure it can detect	Priority
Period consistency	Fact contexts / fiscal periods	Mixing wrong year, quarter, instant vs duration	Highest
Unit consistency	USD, shares, percentages, scale	Comparing or combining incompatible units	Highest
Calculation relationship	XBRL calculation link / curated identity	Components do not reconcile to total	High if linkbase coverage is reliable
Balance-sheet identity	Assets, liabilities, equity	Inconsistent statement values / bad retrieval	High
Segment reconciliation	Dimensional member values + total	Missing segment, duplicate segment, wrong total	Medium; depends on reporting structure
Cross-source consistency	LLM/KG value vs authoritative XBRL fact	Structured factual disagreement	Mostly Stage 4

3.3 Auditor behavior

The first version should not "fix" answers by itself. It should classify the evidence/result and trigger a controlled recovery path.

Audit result	System action
Pass	Allow answer synthesis / finalization
Arithmetic failure	Recompute from retrieved operands; if operands are suspect, re-retrieve
Period/unit failure	Re-retrieve with constrained period/unit filters
Accounting identity failure	Retrieve missing/alternative components and verify again
Unsupported check	Do not pretend it was verified; pass through with an explicit unverified status

3.4 Auditor metrics

- Detection precision: when the auditor flags a case, how often is there a real inconsistency?
- Detection recall on seeded or naturally observed errors.
- False-positive rate on known-correct evidence sets.
- Recovery rate: proportion of flagged cases that become traceably correct after re-retrieval/recomputation.
- End-to-end accuracy delta and traceable-correct delta from graph alone to graph + auditor.
- Latency and token overhead; deterministic checks should be separated from any LLM adjudication cost.

Stage 3 checkpoint

You pass when every supported audit check has unit tests, every check is logged, false positives are measured, and the same evaluation set can be run with the auditor toggled on/off. If the auditor catches almost nothing in natural data, seeded-error experiments are useful diagnostics but cannot replace the natural end-to-end result.

Stage 4 - Use XBRL to audit an LLM-built graph (Weeks 16-18, conditional)

Research question: if LLM-built graph structure is still useful for narrative or non-XBRL relations, can authoritative XBRL constrain the subset of graph facts that should be numerically verifiable?

Conditional stage

Do not begin this until Stages 1-3 have stable results. This stage is valuable only if it tests a new claim rather than delaying the core paper.

4.1 Verification mapping

Map only LLM-graph assertions that have a meaningful XBRL counterpart.

LLM-graph assertion type	Possible XBRL check	Treatment when no check exists
Company-concept-value-period	Direct fact match within normalized context	Leave unverified rather than reject
Total / component relation	Calculation or curated identity	Leave unverified if relationship is not represented
Temporal claim	Compare period-indexed facts	Leave narrative causal claims to text evidence
Acquisition / ownership narrative	Usually not an XBRL fact relation	Do not force-fit into XBRL
Management / risk / strategy relation	No meaningful XBRL verifier	Retain as LLM/text-derived evidence with separate provenance

4.2 Compare three graph conditions

1. **Raw FinReflectKG.** The independent LLM-built baseline.
2. **FinReflectKG + XBRL verification.** Flag or repair only aligned structured assertions.
3. **Native XBRL graph.** The authoritative structured baseline on XBRL-answerable questions.

4.3 What would count as a strong result

- XBRL auditing lowers structured edge/fact error rates in FinReflectKG.
- Questions that depend on corrected/flagged edges improve downstream.
- Narrative coverage is preserved because unverifiable relations are not automatically discarded.
- The combined graph does not simply collapse into the native XBRL graph; it retains complementary non-XBRL evidence.

Stage 5 - Full hybrid financial RAG (Weeks 19-21, conditional)

Research question: how should the system combine authoritative structured retrieval with narrative text retrieval in a realistic end-to-end setting?

5.1 Minimal hybrid architecture

Question class	Primary retrieval path	Verification
Numeric / structural	XBRL graph	Deterministic audit where supported
Narrative	Text RAG	Citation/evidence checks; no fake XBRL verification
Mixed	Both XBRL graph and text RAG	Audit the numeric claims; preserve text provenance for qualitative claims
LLM-KG-specific extension	LLM graph for non-XBRL relations + XBRL constraint layer	Verify only aligned structured assertions

5.2 Router

Measure router accuracy separately. Otherwise, a question sent down the wrong path will look like a retrieval failure. Start with a simple rules/LLM classifier and only make it more complex if errors justify it.

5.3 Do not overbuild the agent loop

- Researcher: chooses path and retrieves evidence.
- Auditor: runs deterministic checks only when applicable.
- Recovery: re-retrieves under tighter structural constraints if a supported check fails.
- Adjudicator/generator: synthesizes the final answer with explicit evidence provenance.
- Stop after one or a small fixed number of recovery iterations so latency remains interpretable.

Stage 5 value

This stage turns the mechanism into a practical system, but it is not necessary to establish the core contribution. If time is short, stop after Stage 3 and write the stronger, narrower paper.

Final experiment design (Weeks 22-23)

6.1 The main accuracy / traceability table

System	Single-fact	Intra-doc	Inter-year	Cross-company	Aggregation	Narrative	Overall
No retrieval							
Flat RAG							
FinReflectKG							
XBRL graph						n/a or limited	
XBRL + auditor						n/a or limited	
Hybrid (if built)							
Oracle / gold evidence							

Report bootstrap confidence intervals in the final paper. If model sampling is stochastic, run multiple seeds for configurations whose variance is material.

6.2 Graph and retrieval table

System	Aligned coverage	Evidence Recall@k	Complete-path recall	Concept-map failure	Wrong-period evidence
Flat RAG				n/a	
FinReflectKG					
XBRL graph					

6.3 Cost table

System	Index build wall time	Index LLM tokens	Index API/LLM cost	Online latency	Online tokens
Flat RAG		0 for graph construction			
FinReflectKG		Use published / measured construction information where comparable			
XBRL graph		0 for XBRL graph construction	Compute/storage only		
XBRL + auditor	same index	0 additional construction tokens	Deterministic audit compute		

6.4 Required ablations

- XBRL graph with auditor off versus on.
- XBRL graph using only direct concept retrieval versus graph traversal.
- Temporal edges off for inter-year questions.
- Cross-company alignment path off for cross-company questions.
- Each major audit check removed individually if the check contributes enough cases to measure.
- Router off (send mixed questions to both paths) if Stage 5 is built.
- XBRL verification off inside FinReflectKG if Stage 4 is built.

6.5 Memorization control

Always retain the no-retrieval baseline and Traceable Correct / Untraceable Correct split. If feasible, add recent filings beyond the model's likely training window or otherwise create a held-out temporal control. The objective is to distinguish retrieval improvements from model recall of famous-company financial facts.

Week-by-week execution plan

The estimates below assume 15-20 hours per week. "Expected progress" is the amount that should be finished by the end of the week, not a promise that every experiment will succeed.

Week	Focus	Est. time	Expected progress	End-of-week gate
1	Scope + evaluation harness	15-18 h	Freeze RQ1-RQ5, overlap definition, common evidence schema, dev/test split, metric functions, cache structure.	One command runs no-retrieval + flat RAG on fixed dev slice.
2	FinReflectKG/XBRL overlap	15-20 h	Inventory FinReflectKG companies/years; map the exact initial overlap; pull matching XBRL facts; record unsupported narrative-only cases.	Aligned company/filing manifest committed to repo.
3	XBRL fact store	16-20 h	Normalize entity, concept, period, unit, dimensions, value, filing provenance. Add fiscal-period and restatement tests.	Fact store passes hand checks on 2-3 companies.
4	XBRL concept graph	16-20 h	Add presentation relationships; establish calculation-link feasibility; create temporal and cross-company alignment edges.	Can inspect/traverse a concept neighborhood end to end.
5	Question-to-concept seeding	15-18 h	Implement alias/concept resolution for the dev questions; log mapping confidence and failures rather than hiding them.	Most XBRL-answerable dev questions resolve to usable seed concepts.
6	Stage 1 retrieval comparison	16-20 h	Run FinReflectKG and XBRL retrieval on aligned dev/test slices; measure evidence recall, coverage, failure categories.	First Stage 1 graph/evidence table exists.
7	Stage 1 QA + cost	16-20 h	Use same generator across graph systems; compute QA/traceability; measure index construction/runtime costs.	Stage 1 go/no-go decision written in one page.
8	Multi-hop benchmark setup	15-18 h	Load/verify FinReflectKG-MultiHop or best available structural set; label questions by hop/relation class; add XBRL-answerability tag.	Evaluation set has structural labels and gold evidence where possible.
9	Multi-hop system runs	16-20 h	Run Flat RAG, FinReflectKG, and XBRL graph; capture full retrieved paths, not just final answers.	1-hop/2-hop/3-hop and relation-type tables populated.
10	Generated XBRL multi-hop set	15-20 h	Generate a larger exact-answer supplement for inter-year/cross-company/ aggregation; hand-check a sample; freeze templates.	Large supplemental set with exact gold evidence exists.
11	Stage 2 analysis	15-18 h	Bootstrap CIs; inspect where graph helps/hurts; write mechanism-level failure notes; decide whether H2 is supported.	Headline multi-hop figure + 5-10 representative cases.
12	Auditor core	16-20 h	Implement endpoint arithmetic, period, and unit checks with unit tests and structured logs.	Auditor can deterministically accept/reject curated cases.

Week	Focus	Est. time	Expected progress	End-of-week gate
13	Accounting checks	16-20 h	Add reliable calculation identities or curated identities; balance-sheet and selected reconciliation checks.	Supported checks have test coverage and explicit unsupported states.
14	Recovery loop	15-20 h	On audit failure, re-retrieve under period/unit/concept constraints; cap retries; record whether recovery succeeds.	Auditor-triggered recovery works end to end on dev cases.
15	Stage 3 evaluation	16-20 h	Run auditor on/off ablation; measure precision/recall, false positives, recovery rate, accuracy/traceability delta.	Stage 3 result is strong enough to keep, narrow, or drop.
16	Stage 4 feasibility gate	12-18 h	Only if core is stable: define XBRL-to-FinReflectKG assertion alignment and estimate how many KG facts are verifiable.	Decision: build Stage 4 or skip it.
17	Audit LLM-built graph	15-20 h	Verify aligned FinReflectKG facts/relations against XBRL; flag disagreement; preserve unverifiable narrative relations.	Raw vs audited FinReflectKG graph-quality comparison.
18	Stage 4 downstream test	15-18 h	Trace whether corrected/flagged graph assertions change retrieval/QA outcomes; stop if effect is negligible.	Stage 4 kept only if it adds a distinct result.
19	Stage 5 router/hybrid	15-20 h	Only if time remains: implement numeric/narrative/mixed routing and combine XBRL with text RAG.	Router accuracy measured separately from retrieval accuracy.
20	Hybrid integration	16-20 h	Integrate provenance-aware evidence synthesis and numeric auditing for mixed questions; avoid unnecessary agent complexity.	End-to-end mixed-question pipeline works on dev slice.
21	Stage 5 ablation / freeze	15-18 h	Compare router vs both-path retrieval; freeze architecture. If hybrid adds little, remove from headline claims.	Final system list for paper is frozen.
22	Final benchmark runs	18-20 h	Run frozen configurations on held-out sets; multiple seeds where needed; collect latency/tokens/cost logs.	No more architecture changes after this week.
23	Statistics + figures	16-20 h	Bootstrap CIs, structural-class plots, traceability tables, graph-quality table, cost table, ablations.	All paper tables/figures exist before prose.
24	Write methods + results	16-20 h	Write from the tables; document data alignment, graph schema, retrieval, auditing, metrics, and negative results.	Complete methods/results draft.

Week	Focus	Est. time	Expected progress	End-of-week gate
25	Intro + related work + limitations	15-18 h	Write thesis carefully; foreground XBRL coverage limits, benchmark size, memorization, fiscal alignment, and generalization limits.	Full paper draft + limitations section.
26	Reproduction + release	18-20 h	Fresh clone/environment run; verify seeds/configs; package code, index builder, audit rules, generated questions, and result scripts.	Clean reproduction and submission-ready artifact.

Formal checkpoints and decisions

Checkpoint	When	Must have	If it fails
A - Infrastructure	End W2	Fixed slices, shared schema, baseline harness, overlap manifest	Do not build more graph features; fix data/evaluation first
B - Stage 1	End W7	Fair XBRL vs FinReflectKG evidence + QA + cost comparison	Diagnose mapping/coverage/traversal; narrow claim if needed
C - Stage 2	End W11	Accuracy/evidence by hop and structural class	If structure does not matter, pivot toward analysis/negative result
D - Stage 3	End W15	Auditor on/off ablation with false positives and recovery metrics	Keep only checks that catch meaningful errors; do not overstate deterministic coverage
E - Extension gate	W16	Stable core paper story	Skip Stages 4-5 if core is not yet stable
F - Architecture freeze	End W21	Final system list and ablations	No more feature additions
G - Submission	End W26	Reproducible final tables from clean environment	Delay claims, not validation

Fallback paths

Observed result	Pivot
XBRL retrieval is not better overall but excels on specific structural classes	Lead with conditional structural advantage; make per-class analysis the main contribution.
XBRL graph is competitive but auditor adds little	Publish retrieval/indexing contribution; keep deterministic audit as a measured negative/limited result.
Auditor is the strongest result	Lead with deterministic verification; treat graph source as secondary.
FinReflectKG edge/fact noise is the clearest finding	Lead with graph-quality/downstream error propagation and XBRL as authoritative reference on aligned facts.

Observed result	Pivot
XBRL coverage is too narrow for natural benchmarks	Use the coverage boundary as a first-class analysis and retain hybrid text retrieval; do not score narrative questions as XBRL failures.
No strong accuracy gain but Untraceable Correct is high	Pivot toward evaluation methodology: financial RAG benchmark scores are confounded by memorization and unsupported correctness.
Stage 4/5 extensions add complexity but little measurable gain	Delete them from the headline system. A narrower paper is stronger than an unablated architecture.

Risks and controls

Risk	Why it matters	Control
Unfair FinReflectKG comparison	Different company/year/concept coverage can create fake wins	Primary results use explicit overlap; report coverage separately
Concept mapping errors	Wrong XBRL seed makes a good graph look bad	Log mappings, confidence, aliases, and unmapped questions; hand-review a sample
Fiscal/calendar-year mismatch	Produces confident but silent wrong comparisons	Period normalization tests; auditor checks; preserve raw contexts
Restatements / duplicate facts	Same concept-period can have multiple valid reported values	Keep accession/provenance and selection policy explicit
Calculation-link coverage is inconsistent	Could make auditor look more general than it is	Measure link coverage; use curated identities as a transparent fallback
150-question statistical weakness	Small differences are unstable	Bootstrap CIs + larger multi-hop/generated supplement + multiple seeds where needed
LLM memorization	Correct answer may not come from retrieved evidence	No-retrieval baseline + Traceable/Untraceable Correct + temporal control if feasible
Stage 4/5 scope creep	Could consume weeks without strengthening the thesis	Conditional extension gate at Week 16
Synthetic benchmark overuse	Templated questions can be easier than natural questions	Use only as supplement; hand-validate; report natural benchmark results separately
Graph construction cost overclaim	Parsing/indexing is not literally free	Say zero LLM construction calls/tokens for XBRL graph; still report wall time/storage

Repository organization and expected artifacts

Keep the implementation structured around experiments rather than around one giant agent pipeline.

Area	Suggested artifact
data/manifests	FinReflectKG overlap manifest; benchmark split manifests; company/year lists
src/xbrl	SEC/XBRL ingestion, normalization, concept mapping, graph construction

Area	Suggested artifact
src/retrieval	Flat retriever, FinReflectKG adapter, XBRL traversal, shared evidence schema
src/audit	Arithmetic, period, unit, accounting identity checks + recovery policy
src/eval	QA metrics, evidence/path metrics, traceability, bootstrap, cost logging
experiments/stage1	XBRL vs FinReflectKG scripts/configs/results
experiments/stage2	Multi-hop structural analysis
experiments/stage3	Auditor ablations and recovery experiments
experiments/stage4	Conditional LLM-KG auditing
experiments/stage5	Conditional hybrid router/system
artifacts	Frozen figures, tables, generated question set, identity-check definitions
configs	Model/retrieval parameters and seeds; no hidden notebook-only settings

What should exist before writing begins

- A frozen README-level thesis and research-question list.
- One reproducible command per stage.
- A manifest for every benchmark/split used in the paper.
- Raw result files plus scripts that regenerate every reported table.
- Audit logs showing which deterministic checks fired and why.
- A failure-case notebook/document containing representative wins and losses for every major system.
- A limitations list maintained during experiments, not invented at the end.

Proposed paper skeleton

Section	What it should argue
1. Introduction	Financial RAG repeatedly reconstructs structure that SEC XBRL already encodes; authoritative structure may help both retrieval and verification.
2. Related work	Flat financial RAG; GraphRAG; FinReflectKG; multi-hop financial QA; XBRL as machine-readable reporting structure; verification/auditing.
3. Problem setup	Question structural taxonomy, XBRL-answerability boundary, evidence/traceability definitions.
4. XBRL graph	Fact schema, concept/presentation/calculation structure, temporal/cross-company edges, traversal.
5. Deterministic auditor	Supported checks, recovery behavior, explicit unsupported cases.
6. Experiments	Baselines, aligned datasets, multi-hop split, metrics, cost, statistics, memorization controls.
7. Results	Stage 1, Stage 2, Stage 3; Stage 4/5 only if they earn their place.

Section	What it should argue
8. Analysis	Failure cases, coverage limits, graph-error propagation, traceable vs untraceable correctness.
9. Limitations	Narrative coverage, non-GAAP inconsistency, taxonomy/extension-tag mapping, benchmark scale, generalization to other jurisdictions.
10. Conclusion	Authoritative structure is most useful when used where it is authoritative, rather than pretending one graph representation should answer everything.

Anchor papers and resources from the original plan

These remain the conceptual anchors for the revised plan; the new schedule changes how they are organized, not why they matter.

- LegalGraphRAG - architecture inspiration for Researcher / Auditor / Adjudicator style closed-loop retrieval.
- FinReflectKG - independent LLM-built financial knowledge-graph baseline and graph-quality reference.
- FinReflectKG-MultiHop - structural / multi-hop financial QA benchmark and relation-type stress test.
- Decomposing Retrieval Failures in RAG for Long-Document Financial Question Answering - flat/page/chunk retrieval diagnosis and oracle-style framing for FinanceBench.
- FinanceBench open subset - natural expert-written financial QA benchmark, but too small to support fine-grained claims alone.
- REFinD - possible ground truth for relation-quality analysis on alignable relation classes.
- SEC XBRL Company Facts / Frames / filing XBRL and linkbases - authoritative structured data sources for facts, temporal/cross-company access, and calculation relationships.

One-sentence project summary

> **First test whether XBRL can serve as the financial graph; then stress it with multi-hop questions; then test whether the same authoritative structure can audit reasoning; only after those results are stable should you test XBRL as a constraint on LLM-built graphs or build the full hybrid system.**